

Simple Game

Evirir and Rhae are playing a game. Initially, there are n pairs of integers $(a_1, b_1), \dots, (a_n, b_n)$ and both players have 0 points. Evirir and Rhae take turns to act, with Evirir going first.

On Evirir's turn, they will choose a pair (a_i, b_i) that has not been removed, add a_i to Evirir's total score, and remove the pair (a_i, b_i) .

On Rhae's turn, they will choose a pair (a_i, b_i) that has not been removed, add b_i to Rhae's total score, and remove the pair (a_i, b_i) .

The game ends when all pairs have been removed. Let X and Y be the final score of Evirir and Rhae, respectively. Evirir wants to maximize the value of $X - Y$ and Rhae wants to minimize it. If both players play optimally, what is the final value of $X - Y$?

Input Format

The first line contains one integer, n ($2 \leq n \leq 10^4$), the total number of pairs.

The next n lines contains two space-separated integers, a_i and b_i ($-10^9 \leq a_i, b_i \leq 10^9$), the values of i^{th} pair.

Output Format

Output a single integer, the final value of $X - Y$ if both players play optimally.

Sample Input

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2
4 2
1 3
```

Sample Output

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1
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Sample Explanation

There are two pairs of integers $(4, 2)$ and $(1, 3)$.

There are only two possible cases: - Evirir takes the first pair of integers and Rhae takes the second pair of integers. The final scores: X is 4 and Y is 3, resulting $X - Y = 1$. - Evirir takes the second pair of integers and Rhae takes the first pair of integers. The final scores: X is 1 and Y is 2, resulting $X - Y = -1$.

Since Evirir wants to maximize $X - Y$ and Evirir has the ability to force Rhae to take the remaining pair, Evirir takes the first pair and Rhae takes the second pair, resulting $X - Y = 1$.